

SECSR2043 OPERATING SYSTEMS

[20 Marks]

Name : Yasmin Batrisyia Binti Zahiruddin
 Student : A23CS0201
 ID : _____
 Section : 02

Marks

Instruction: Please answer all of the following questions. Whenever the 🙋 symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname.sh* (Example: *ahmad.sh*).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*. *; cp ../fruits/*. *;
cp ../*.jar .
```

- a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **\$HOME** directory.

[4 marks]



Print screen the script that you type;

```
yasmin1@secr2043:~$ nano yasmin.sh
```

```

yasmin1@secr2043: ~
GNU nano 7.2 yasmin.sh
echo ^^|Hello world ^^} > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo ^^|Honda Accord ^^} > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt > orange.txt
cd ../drinks; cp ../cars/*.c .; cp ../fruits/*.c .;
cp ../*.jar .

```

Then draw the tree

```

yasmin1@secr2043:~$ tree
.
|-- cars
|   |-- accord.c
|   |-- civic.c
|   `-- proton.c
|-- dates
|   |-- appointment
|   `-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   `-- proton.c
|-- fruits
|   |-- apple.txt
|   `-- orange.txt
|-- helloworld.jar
|-- mainprocess
|-- mainprocess.c
|-- os_lab
|   |-- echo
|   |-- echo.c
|   |-- hello
|   |-- hello.c
|   |-- sum
|   `-- sum.c
|-- q10
|-- q10.c
|-- subprocess1
|-- subprocess1.c
|-- subprocess2
|-- subprocess2.c
`-- yasmin.sh

```

- b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter "c" as the input to the bash script. [4 marks]



Print screen the bash script you type and run

```
yasmin1@secr2043: ~  
GNU nano 7.2 file_extension.sh  
#!/bin/bash  
read -p "Enter file extension (without dot): " ext  
echo "Files with .$ext extension:"  
find . -type f -name ".*$ext"  
count=$(find . -type f -name ".*$ext" | wc -l)  
echo "Number of files with .$ext extension: $count"
```

```
yasmin1@secr2043:~$ chmod +x file_extension.sh  
yasmin1@secr2043:~$ ./file_extension.sh  
Enter file extension (without dot): c  
Files with .c extension:  
./os_lab/echo.c  
./os_lab/sum.c  
./os_lab/hello.c  
./drinks/civic.c  
./drinks/accord.c  
./drinks/proton.c  
./subprocess1.c  
./subprocess2.c  
./cars/civic.c  
./cars/accord.c  
./cars/proton.c  
./q10.c  
./mainprocess.c  
Number of files with .c extension: 13
```



2. The following Figure 1 illustrates a tree structure of some directories and files.

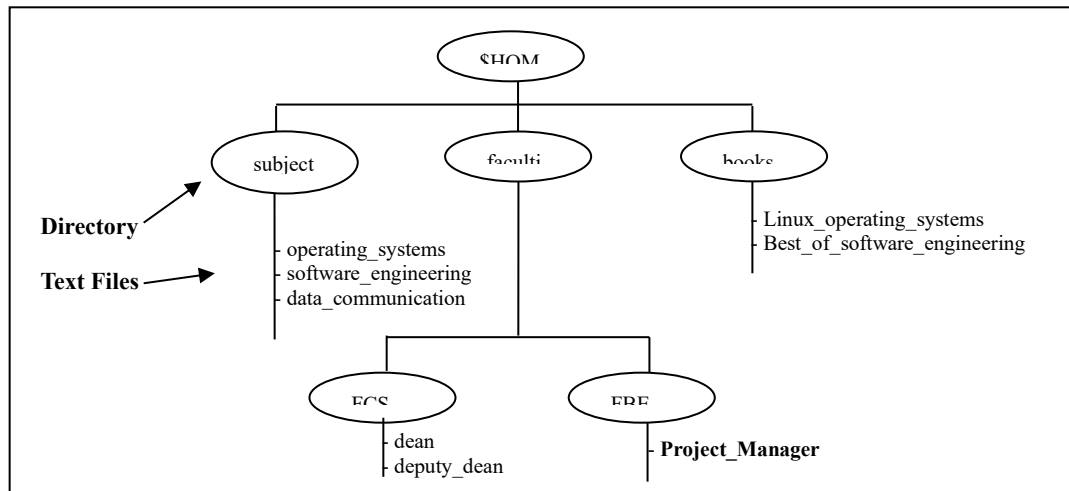


Figure 1

- a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

Print screen the bash script you type and run

```

yasmin1@secr2043: ~
GNU nano 7.2 yasmin2a.sh
#!/bin/bash

# Create the base directory
mkdir -p "$HOME/subject"
mkdir -p "$HOME/facultie/FCS"
mkdir -p "$HOME/facultie/FBE"
mkdir -p "$HOME/books"

# Create text files under 'subject' directory
echo "operating systems" > "$HOME/subject/operating_systems"
echo "software engineering" > "$HOME/subject/software_engineering"
echo "data communication" > "$HOME/subject/data_communication"

# Create text files under 'books' directory
echo "Linux operating systems" > "$HOME/books/Linux_operating_systems"
echo "Best of software engineering" > "$HOME/books/Best_of_software_e

# Create text files under 'FCS' directory
echo "dean" > "$HOME/facultie/FCS/dean"
echo "deputy dean" > "$HOME/facultie/FCS/deputy_dean"

# Create text file under 'FBE' directory
echo "Project Manager" > "$HOME/facultie/FBE/Project_Manager"

echo "Directory and file structure created successfully."

yasmin1@secr2043:~$ chmod +x yasmin2a.sh
yasmin1@secr2043:~$ ./yasmin2a.sh
Directory and file structure created successfully.
yasmin1@secr2043:~$ tree $HOME
/home/yasmin1
|-- books
|   |-- Best_of_software_engineering
|   |-- Linux_operating_systems
|-- facultie
|   |-- FBE
|   |   |-- Project_Manager
|   |-- FCS
|   |   |-- dean
|   |   |-- deputy_dean
|-- subject
|   |-- data_communication
|   |-- operating_systems
|   |-- software_engineering

```



- b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called book.

[2 marks]

Directory/File	Access Control
books	drwxrwxr-x
subjects	drwxr-xr-x
Best_of_software_engineering	-rw-r--r--
FCS	drwxr-xr-x
project_manager	-rw-r--r--



```

yasmin1@secr2043:~$ cd ~
yasmin1@secr2043:~$ ls -ld books
drwxr-xr-x 2 yasmin1 yasmin1 4096 Jun 29 13:25 books
yasmin1@secr2043:~$ ls -l books
total 8
-rw-r--r-- 1 yasmin1 yasmin1 29 Jun 29 13:25 Best_of_software_engineering
-rw-r--r-- 1 yasmin1 yasmin1 24 Jun 29 13:25 Linux_operating_systems

```

```

yasmin1@secr2043:~$ ls -ld subject
drwxr-xr-x 2 yasmin1 yasmin1 4096 Jun 29 13:25 subject

```

```

yasmin1@secr2043:~$ ls -ld facultie/FCS
drwxr-xr-x 2 yasmin1 yasmin1 4096 Jun 29 13:25 facultie/FCS
yasmin1@secr2043:~$ ls -ld facultie/FBE/Project_Manager
-rw-r--r-- 1 yasmin1 yasmin1 16 Jun 29 13:25 facultie/FBE/Project_Manager

```

- c) Write another bash script (called `myname2c.sh`) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
subjects	✓	✓	✓	✓	x	x	✓	x	x
Best_of_software_engineering	✓	x	✓	x	✓	x	x	x	x
FCS	✓	✓	x	x	x	x	✓	✓	✓
project_manager	x	x	x	x	✓	✓	x	x	✓

Print screen the bash script you type and run

```

yasmin1@secr2043: ~
GNU nano 7.2 yasmin2c.sh
#!/bin/bash

# Set permission for subject directory
chmod 744 $HOME/subject

# Set permission for Best_of_software_engineering file
chmod 520 $HOME/books/Best_of_software_engineering

# Set permission for Best_of_software_engineering file
chmod 607 $HOME/facultie/FCS

# Set permission for project_manager_file
chmod 021 $HOME/facultie/FBE/Project_Manager

yasmin1@secr2043:~$ nano yasmin2c.sh
yasmin1@secr2043:~$ bash yasmin2c.sh

```



- d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
subjects	drwxr--r--
Best_of_software_engineering	-r-x-w----
FCS	drw----rwx
project_manager	-----w---x

```

yasmin1@secr2043:~$ ls -ld subject
drwxr--r-- 2 yasmin1 yasmin1 4096 Jun 29 13:25 subject
yasmin1@secr2043:~$ ls -l books/Best_of_software_engineering
-r-x-w---- 1 yasmin1 yasmin1 29 Jun 29 13:25 books/Best_of_software_engineeri
ng
yasmin1@secr2043:~$ ls -ld faculties/FCS
ls: cannot access 'faculties/FCS': No such file or directory
yasmin1@secr2043:~$ ls -ld facultie/FCS
drw----rwx 2 yasmin1 yasmin1 4096 Jun 29 13:25 facultie/FCS
yasmin1@secr2043:~$ ls -l facultie/FBE/Project_Manager
-----w---x 1 yasmin1 yasmin1 16 Jun 29 13:25 facultie/FBE/Project_Manager

```

End of Lab 3

**** All the Best for Final Exam ****